

CHONGRAN ZHAO

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Education



Brown University

Ph.D. of Engineering in Mechanics of Solids and Structures

Providence, RI

09/2026 – Present



Southern University of Science and Technology

Master of Engineering in Mechanics

Shenzhen, Guangdong

09/2023 – 06/2026



Southern University of Science and Technology

Bachelor of Science in Theoretical and Applied Mechanics

Shenzhen, Guangdong

09/2019 – 06/2023

Research Interests

Continuum Mechanics · Solid Mechanics · Viscoelasticity · Constitutive Modeling · Computational Modeling · Numerical Methods · Soft Materials · Growth and Remodeling

Publications and Manuscripts

- [1] J. Liu, J. Guan, C. Zhao, and J. Luo. A continuum and computational framework for viscoelastodynamics: III. A nonlinear theory. *Computer Methods in Applied Mechanics and Engineering* 430 (2024), 117248.
- [2] J. Liu, C. Zhao, and J. Guan. Modeling finite viscoelasticity based on the Green–Naghdi kinematic assumption and generalized strains. *Journal of the Mechanics and Physics of Solids* 206 (2026), 106346.
- [3] C. Zhao, H. Yuan, and J. Liu. A framework for finite-strain viscoelasticity based on rheological representations. *International Journal of Solids and Structures* 330 (2026), 113893.

Conference

The Chinese Congress of Theoretical and Applied Mechanics

Oral Presentation

A finite viscoelastic constitutive theory based on Green-Naghdi decomposition: A rheological model study

Changsha, China

Jul. 18-21, 2025

18th International Conference on Computational Plasticity

Oral Presentation

A rational framework of finite viscoelasticity: Dissipation potential and non-Newtonian effects

Barcelona, Spain

Sep. 2-5, 2025

Summer School on Physics-Informed Modeling, Simulation and Experiments with Emphasis on the Cardiovascular System

Participant

Graz, Austria

Sep. 15-19, 2025

Skills

- **Technical Skills:** C++ · CMake · Matlab · Linux/Unix · Git · L^AT_EX
- **Languages:** Chinese · English